FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – IV
PAPER –2
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

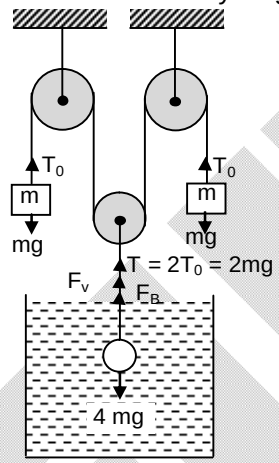
Physics

PART – I

SECTION – A

1. B

Sol. From the free body diagram of the sphere:



$F_v = 4\,mg - 2\,mg - F_B$

$\Rightarrow \quad F_v = 2\,mg - F_B$

$\Rightarrow \quad 6\pi\eta r v = \frac{3}{4}\pi r^3\left(\frac{\sigma}{2} - \rho\right)g$

(Since, $4m = \frac{4}{3}\pi r^3 \times \sigma$)

$\Rightarrow \quad v = \frac{2}{9}r^2\frac{(\sigma - 2\rho)g}{\eta}$

2. A

Sol. $\sigma \propto \frac{1}{R}, \frac{\sigma_2}{\sigma_1} = \frac{3}{4} = \underbrace{\tan\theta}_{\text{slope}}$

3. C

Sol. By right hand thumb rule; the field at P* due to both the segments is inside the plane of the paper, i.e., along negative z-axis.

4. A

Sol. $i_3 = i_1 + i_2 = 5\left(\frac{3}{5}\sin\omega t + \frac{4}{5}\cos\omega t\right)$

$= 5\left(\sin\omega t \cdot \cos 53^\circ + \cos\omega t \cdot \sin 53^\circ\right)$

$= 5\sin\left(\omega t + 53^\circ\right)$

5. AD

Sol. Before collision

$\omega_P = \sqrt{\frac{k}{m}} \qquad \omega_Q = \sqrt{\frac{k}{m}}$

After collision

$\omega_{combined} = \sqrt{\frac{3k}{3m}} = \sqrt{\frac{k}{m}}$

Also, conserving momentum before and after collision

$(2m)(2A\omega) - (m)(A)\omega = (3m)A'\omega$

$A' = A$

6. BC

Sol. $x = 3t^2 - t^3$

At $t = 0, x = 0$

$t = 4, x = 48 - 64 = -16$

$\Delta x = X_b - X_i = -16$

$\Delta x = X_b - X_i = -16$

$V = 6t - 3t^2$

$V = 0, t = 2$ sec

$t = 2, X = 4$

$t = 2, X = 4$

Distance = $4 + 4 + 16 = 24$ m

7. BCD

Sol. At steady state, no current passes through capacitor and voltmeter.

$i = \frac{2}{20} = \frac{1}{10}$ A

Potential difference across 1 μF is $\frac{1}{10} \times 8 = 0.8$ volt

SECTION – B

8. 45

Sol. $F_{net} = 3N$

$a_{CM} = \frac{3}{m}; \tau_{net} = 7r \qquad \ldots(i)$

$\alpha_{net} = \frac{7r}{\frac{mr^2}{2}} = \frac{14}{mr}$

$$\pi = \frac{1}{2}\alpha t^2 \qquad \ldots(ii)$$

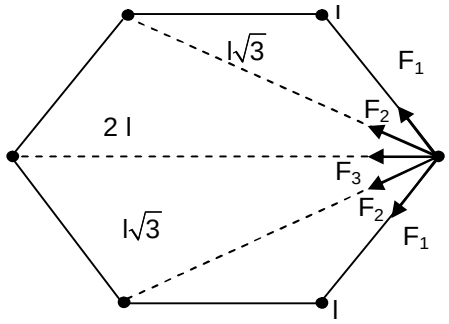
$$t = \sqrt{\frac{\pi m r}{7}}$$

$$v_{CM} = a_{CM} t = \frac{3}{m}\sqrt{\frac{\pi m r}{7}}$$

$$(v_{CM})^2 = 45 \text{ m/s}$$

9. 2

Sol.



$$F_1 = \frac{Gm^2}{l^2}, F_2 = \frac{GM^2}{\left(l\sqrt{3}^2\right)}, F_3 = \frac{Gm^2}{(2l)^2}$$

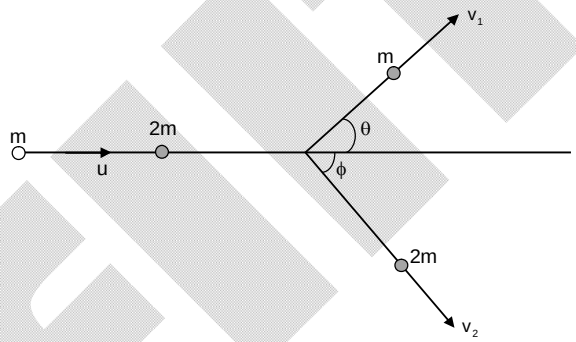
$$F_{res} = 2F_1\cos 60^\circ + 2F_2\cos 30^\circ + F_3$$

$$= \frac{Gm^2}{l^2}\left(1 + \frac{1}{\sqrt{3}} + \frac{1}{4}\right)$$

$x = 1, y = 3$ and $z = 4$

10. 90

Sol.



$$mu = mv_1\cos\theta + 2mv_2\cos\phi \quad \ldots. (i)$$

$$mv_1\sin\theta = 2mv_2\sin\phi \quad \ldots. (ii)$$

$$\frac{2}{3}\left(\frac{1}{2}mu^2\right) = \frac{1}{2}2mv_2^2 \quad \ldots. (iii)$$

$$\frac{1}{3}\left(\frac{1}{2}mu^2\right) = \frac{1}{2}mv_1^2 \quad \ldots. (iv)$$

Solving equation, $v_1 = v_2 = \frac{u}{\sqrt{3}}$

$\theta + \phi = 120^\circ$ and $\theta = 90^\circ$

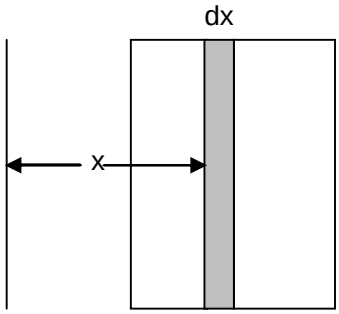
11. 4

Sol. After moving the loop about side PQ through 120°, the normal on the side PS from the wire will be passing through midpoint C and the net magnetic flux through the loop in the new position will be zero.

$\therefore$ Change in flux $\Delta f =$ Flux through the loop in the initial through area element dA

$$d\phi = \vec{B}\cdot d\vec{A} = \frac{\mu_0 I}{2\pi x} a \cdot dx$$

$$\phi = \int_a^{2a} \frac{\mu_0 I a dx}{2\pi x} = \frac{\mu_0 I a}{2\pi} \ln(2)$$



$\therefore$ Charge flown through the square loop

$$= \frac{\Delta\phi}{R} = \frac{\mu_0 I a}{2\pi R} \ln(2)$$

12. 180

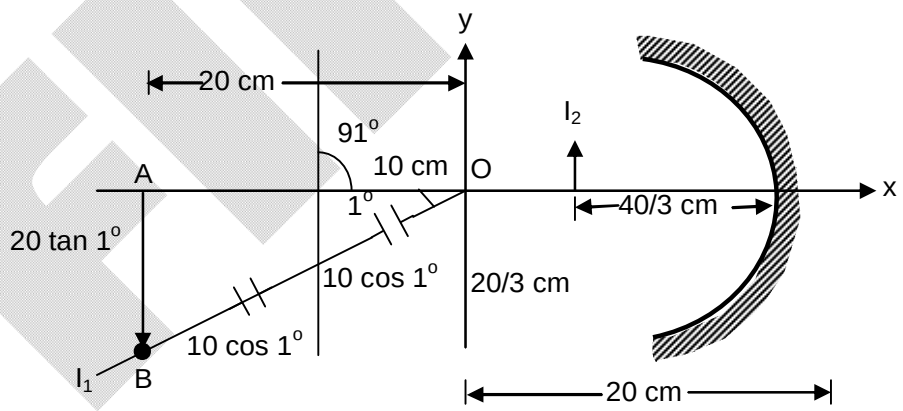
Sol. Distance of I_1 from mirror = 40 cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \quad \frac{1}{-10} = \frac{1}{v} + \frac{1}{-140}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{40} - \frac{1}{10}$$

$$\Rightarrow v = -\frac{40}{3} \text{cm}$$



Using magnification formula

$$m = \frac{I}{O} = -\frac{v}{u}$$

$$\Rightarrow \quad \frac{I}{-20\tan 1^{o}} = -\frac{-\frac{40}{3}}{-40}$$

$$\Rightarrow \quad I = \frac{20\tan 1^{o}}{3}$$

$$= \frac{20}{3}\tan\frac{\pi}{180} \simeq \frac{20}{3}\times\frac{\pi}{180} = \frac{\pi}{27}$$

Co-ordinate of image = $\left(\frac{20}{3}, \frac{\pi}{27}\right)$

13. 2

Sol. $E = \frac{Z^2}{n^2}$

In series limit of bracket axis, transition is for $n = \infty$ to n = 4
In series limit of Balmer series, transition for $n = \infty$ to n = 2

$$hv_{bracket} = 13.6Z^2\left(\frac{1}{16}\right)$$

$$hv_{Balmer} = 13.6\left(\frac{1}{4}\right)$$

$$\frac{13.6Z^2}{16} = \frac{13.6}{4} = Z = 2$$

SECTION – C

14. 6.67

Sol. Linear impulse momentum theorem

$$\int F dt = \Delta p$$

$$\left[\frac{1}{2}\times 4\times 600 + 600(\Delta t)\right]\times 2 - 150\times 10\times \sin 30^{o}\left[4+\Delta t\right]$$

$$= 150\left[0-(-4)\right]$$

$$\Delta t = 2.67\,\text{sec.}$$

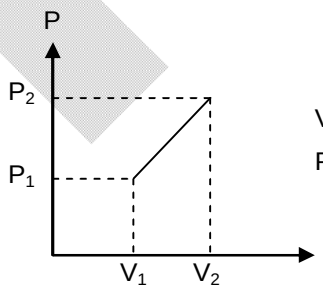
$$t = 4 + \Delta t = 6.67\,\text{sec.}$$

15. 4.00

Sol. Apply linear impulse moment theorem as in above case.

16. 1.50

Sol. W = Area under the curve = $\frac{3}{2}P_1V_1$ and $P_1V_1 = nRT_1$



$V_2 = 2V_1$

$P_2 = 2P_1$

Therefore $\frac{w}{nRT_1} = \frac{\frac{3}{2} \cdot P_1V_1}{P_1V_1}$

17. 6.00

Sol. $Q = dU + W$

$dU = nC_v dT$

For final state $P_2V_2 = 2P_1 2V_1$

$= 4P_1V_1 = nR(4T_1)$

Hence final temperature is $4T_1$

$dU = n \cdot \frac{3}{2} R \cdot 3T_1 = \frac{9}{2} nRT_1$

$Q = \frac{3}{2} \cdot nRT_1 + \frac{9}{2} nRT_1 = 6nRT_1$

$\frac{Q}{nRT_1} = 6$

Chemistry

PART – II

SECTION – A

18. C
Sol. Due ring opening formation of C takes place.

19. B
Sol. Due to stability of carbocation.

20. D
Sol. Due to formation of carbocation.

21. A
Sol. Due to ring opening.

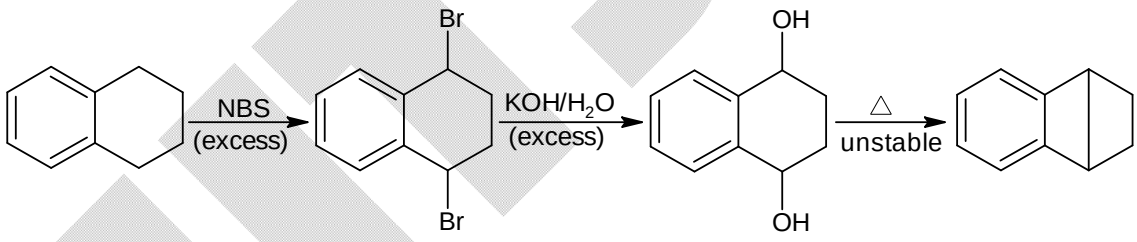
22. ABC
Sol. All the symmetrical molecules have zero dipole.

23. BD
Sol. Facts

24. AC
Sol. 2° Amine does not dissolve in an alkali after reaction with Hinsberg's reagent.

SECTION – B

25. 6
Sol.



26. 1
Sol. $E = E^{o} - \frac{2.303RT}{nF}\log\frac{[Product]}{[Reactant]}$

27. 4
Sol. From COOH, SH, Amide, Alkyne.

28. 1
Sol.

X	+	Y	⇌	2Z
1		1.5		0.9
(1-x)		(1.5-x)		0.5 + 2x

Given 0.5 + 2x = 1
2x = 0.5
$x = \frac{0.5}{20} = \frac{1}{4}$

$$K = \frac{(0.5+2x)^2}{(1-x)(1.5-x)}$$

$$= \frac{(1)^2}{(0.75)(1.25)} = \frac{14}{15} = 1.06$$

29. 6

Sol. Use the formula.

$$K = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$$

30. 7

Sol. Because $7\alpha - H$ are present is compound.

SECTION – C

31. 3.58

Sol. $\Delta T_f = i.K_f.m$ $\quad HCOOH \rightleftharpoons HCOO^- + H^+$

$0.109(1-\alpha) \quad 0.109\alpha \quad 0.109\alpha$

Therefore,

$\alpha = 0.0358$

32. 1.45

Sol. $K = \frac{0.109\alpha \times 0.109\alpha}{0.109(1-\alpha)} = 1.45 \times 10^{-4}$

33. 57.13

Sol. $P_T = P_A^o X_A + P_B^o X_B$

34. 0.60

Sol. $\frac{P - P^o}{P^o} = \chi_\beta$

Mathematics

PART – III

SECTION – A

35. A

Sol. $x^2ydy+\left(x^3+x^2y\right)dx=2xy^2dx+y^3dx$

$\Rightarrow x^2ydy-y^2xdx+x^3dx+x^2ydx=xy^2dx+y^3dx$

$\Rightarrow xy\left(xdy-ydx\right)+x^2\left(x+y\right)dx=y^2\left(x+y\right)dx$

$\Rightarrow \int\frac{\frac{y}{x}d\left(\frac{y}{x}\right)}{\left(\frac{y^2}{x^2}-1\right)\left(\frac{y}{x}+1\right)}=\int\frac{dx}{x}$

$\frac{y}{x}=t$

$\Rightarrow \int\frac{tdt}{\left(t^2-1\right)\left(t+1\right)}=\int\frac{dx}{x}$

$\frac{1}{4}\int\frac{dt}{t-1}-\frac{1}{4}\int\frac{dt}{t+1}+\frac{1}{2}\int\frac{dt}{\left(t+1\right)^2}=\log x+c$

$\Rightarrow \log\left|\frac{t-1}{t+1}\right|=\frac{2}{t+1}+\log x^4+k$

36. A

Sol. $g'\left(f\left(x\right)\right)=\frac{1}{f'\left(x\right)}\Rightarrow g'\left(x^3+4x-1\right)=\frac{1}{3x^2+4}$

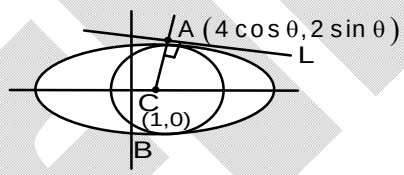
$\Rightarrow g''\left(x^3+4x-1\right)=\frac{-6x}{\left(3x^2+4x\right)^3}$

put x = 1

$\Rightarrow g''\left(4\right)=\frac{-6}{343}$

37. C

Sol.



Equation of tangent $L:\cos\theta+2y\sin\theta=4$

$m_{AC}.m_L=-1$

$\Rightarrow \cos\theta=\frac{1}{3}$

$\Rightarrow A\left(\frac{4}{3},\frac{4\sqrt{2}}{3}\right)$

$\Rightarrow R_{max}=CA=\sqrt{\frac{11}{3}}$

$\Rightarrow \dfrac{A}{S} = \dfrac{\pi ab}{\pi R_{max}^2} = \dfrac{4\times 2\times 3}{11} = \dfrac{24}{11}$ sq. units

38. B

Sol. System is consistent

$\Rightarrow \begin{vmatrix} k & 1 & k+1 \\ 2 & 1 & 2k-1 \\ 1 & -k & 3 \end{vmatrix} = 0$

$\Rightarrow k(3+2k^2-k)-1(6+1-2k)+(k+1)(-2k-1)=0$

$\Rightarrow 2k^3-3k^2+2k-8=0$

let $f(k) = 2k^3-2k^2+2k-8$

$f'(k) = 2(3k^2-3k+1) > 0 \ \forall k \in R$

$\Rightarrow$ cubic has only one real root (further for this value of k, no two lines are parallel)

39. CD

Sol. $x^2-(a^2+a+1)x+a(a^2+1)=0$

Its greater root is a^2+1

Let $f(x) = x^2-a^2x-2(a^2-2)=0$

$\therefore f(a^2+1) < 0 \Rightarrow a^2 > 5$

Least positive integral value of a is 3

40. ABC

Sol. Let f takes its largest value at some $c \in (0, 4) \therefore f'(c) = 0$

Now, in $y = f'(x)$ using LMVT in [0, c] and [c, 4]

$\therefore |f''(d)| = \left|\dfrac{f'(c)-f'(0)}{c-0}\right| \le 5$

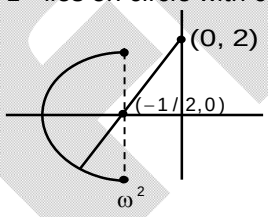
Again, $|f''(d)| = \left|\dfrac{f'(4)-f'(c)}{4-c}\right| \le 5$

$\Rightarrow |f'(4)| \le 5(4-c)$

41. BC

Sol. $\arg\left(\dfrac{z^2-\omega}{z^2-\omega^2}\right) = \dfrac{\pi}{2}$

z^2 lies on circle with centre



$\left(-\dfrac{1}{2},0\right)$ and $\dfrac{\sqrt{3}}{2}$ units

$|z^2|_{max} = \dfrac{1}{2}+\dfrac{\sqrt{3}}{2}$

$|z|_{max} = \sqrt{\dfrac{1+\sqrt{3}}{2}} = \dfrac{\sqrt{2+2\sqrt{3}}}{2}$

$|z-1-i|\,|z+1+i|_{max}$

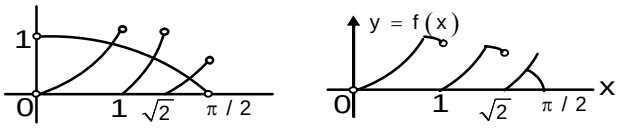
$|z^2-2i|_{max}=\frac{\sqrt{17}}{2}+\frac{\sqrt{3}}{2}$

SECTION – B

42. 7

Sol. $f(x)=\min\left(\{x^2\},\cos x\right);x\in\left(0,\frac{\pi}{2}\right)$

$$\{x^2\}=\begin{cases}x^2 & 0\le x<1\\ x^2-1 & 1\le x<\sqrt{2}\\ x^2-2 & \sqrt{2}\le x<\frac{\pi}{2}\end{cases}$$



Discontinuous at 2 points = m
Non – differential 5 points = n
m + n = 7

43. 6

Sol. $S=\sum_{k=0}^{\infty}\left(\frac{2^k}{7^{2^k}+1}\right)$

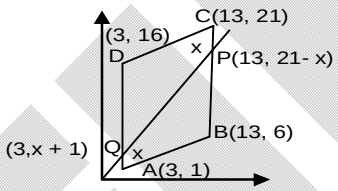
$S=\frac{2^0}{7^1+1}+\frac{2^1}{7^2+1}+\frac{2^2}{7^4+1}+\frac{2^3}{7^8+1}$

$\left(\frac{1}{7-1}-\frac{2}{7^2-1}\right)+\left(\frac{2}{7^2-1}-\frac{2^2}{7^4-1}\right)+\left(\frac{2^2}{7^4-1}-\frac{2^3}{7^8-1}\right)+....$

$S=\frac{1}{6}\Rightarrow\frac{1}{5}=6$

44. 11

Sol.



For congruent let AQ = CP = x

$\Rightarrow m=\frac{21-x}{13}=\frac{x+1}{3}\Rightarrow x=\frac{25}{8}$

$\therefore m=\frac{1}{3}\left(\frac{25}{8}+1\right)=\frac{11}{8}$

45. 10

Sol. $\tan9\theta=\frac{3}{4}\Rightarrow\sin9\theta=\frac{3}{5},\cos9\theta=\frac{4}{5}$

$\therefore\frac{3}{\sin3\theta}-\frac{4}{\cos3\theta}=10\left(\frac{\sin(9\theta-3\theta)}{\sin6\theta}\right)=10$

46. 48

Sol. $I = \int_1^5 f(x)dx + \int_2^{10} f^{-1}(y)dy$

Let $f^{-1}(y) = t \Rightarrow dy = f'(t)dt$

$\Rightarrow I = \int_1^5 f(x)dx + \int_1^5 tf'(t)dt$

$\Rightarrow I = \int_1^5 \left(f(x) + xf'(x)\right)dx = 48$

47. 7

Sol. Let the numbers be 1, 2, 3, ….., n and the erased number be x

$\Rightarrow \frac{\frac{n(n+1)}{2} - n}{n-1} \le 35\frac{7}{17} \le \frac{\frac{n(n+1)}{2} - 1}{n-1}$

$\frac{n}{2} \le \frac{602}{17} \le \frac{n+2}{2} \Rightarrow 68.8 \le n \le 70.8$

Hence,

If n = 69 then $\frac{\frac{69\times 70}{2} - x}{68} = \frac{602}{17} \Rightarrow x = 7$

If $n = 70$ then $x \notin I$.

SECTION – C

48. 1.41

49. 12.00

Sol. Sol. Mid-point (–1, –1, 2) lies on plane P. Direction ratios of normal to plane P is < –3, 3,–1 >

∴ Plane P is $3x - 3y + z - 2 = 0$

$\therefore \vec{n}_1 = 3\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{n}_2 = \hat{i} - \hat{j} + \lambda\hat{k}$

$\Rightarrow \cos 45^0 = \left|\frac{3+3+\lambda}{\sqrt{19}\sqrt{2+\lambda^2}}\right|$

$\Rightarrow 17\lambda^2 - 24\lambda - 34 = 0$

sum of values of $\lambda = \frac{24}{17} = 1.41$

Point on line L is $(1+k, 3k-2, 7-k)$

$\Rightarrow k = 2 \Rightarrow R(x_0, y_0, z_0) = (3,4,5)$

50. 6.00

51. 1.86

Sol. $f(x) = e^x(2x-1) - ax + a$

$\Rightarrow f(0) = a - 1 < 0$ and $f(1) = e > 0$

Also $f'(x) = e^x(2x+1) - a$

$\Rightarrow f(x)$ is increasing when x > 1 and f(x) is decreasing when x < -1

Now $f(x_0) < 0$ for only one $x_0 \in I$.

$\therefore f(-1) \ge 0 \Rightarrow a \in \left[\frac{3}{2e}, 1\right)$

$\Rightarrow p = 3, q = 2, r = 1$